過去問2

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1

(1) $x''-5x'+6x=0$

$\lambda^2-5\lambda+6=0$

$(\lambda-3)(\lambda-2)=0$

$\lambda=3,2$

$\therefore x=C_1e^{3t}+C_2e^{2t}$

(2) $x''-4x'+4x=0$

$\lambda^2-4\lambda+4=0$

$(\lambda-2)^2=0$

$\lambda=2$ (2重解)

$x=(C_1+C_2)e^{2t}$

(3) $x''-4x'+13x=0$

$\lambda^2-4\lambda+13=0$

$\lambda=\frac{4\pm\sqrt{16-4\cdot13}}{2}=\frac{4\pm\sqrt{-36}}{2}$

$=\frac{4\pm\sqrt{36}i}{2}$

$=\frac{4\pm6i}{2}$

$=2\pm3i$

$\therefore x=e^{2t}(C_1\cos3t+C_2\sin3t)$

(4) $x''-4x'=0$

$\lambda^2-4\lambda=0$

$\lambda(\lambda-4)=0$

$\lambda=0,4$

$x=C_1+C_2e^{4t}$

(5) $x''-4x=0$

$\lambda^2-4=0$

$(\lambda+2)(\lambda-2)=0$

$\lambda=\pm2$

$\therefore x=C_1e^{2t}+C_2e^{-2t}$

(6) $x''+4x=0$

$\lambda^2+4=0$

$\lambda^2=-4$

$\lambda=2i$

$\therefore x=C_1\cos2t+C_2\sin2t$

(7) $x''-5x'+6x=6t+7$

$\lambda^2-5\lambda+6=0$

$x_0=C_1e^{3t}+C_2e^{2t}$

$x_1=At^2+Bt+C$

$x_1'=2At+B,\ x_1''=2A$

$2A-5(2At+B)+6(At^2+Bt+C)$

$2A-10At+5B+6At^2+6Bt+6C$

$6At^2+t(6B-10A)+2A+5B+6C$

$6A=0$

$\begin{cases}6B-10A=6\\2A+5B+6C=7\end{cases}$

$\therefore\begin{cases}A=0\\B=1\\C=\frac{1}{3}\end{cases}$

$\therefore x=x_1+x_0$

$=C_1e^{3t}+C_2e^{2t}+t+\frac{1}{3}$

(8) $x''-4x'+13x=9e^{2t}$

$x_0=e^{2t}(C_1\cos3t+C_2\sin3t)$

$x_1=Ae^{2t},\ x_1'=2Ae^{2t},\ x_1''=4Ae^{2t}$

$\therefore 4Ae^{2t}-8Ae^{2t}+13Ae^{2t}=9e^{2t}$

$9Ae^{2t}=9e^{2t},\ A=1$

$x=x_0+x_1=e^{2t}(C_1\cos3t+C_2\sin3t)+e^{2t}$

(9) $x''-4x'+4x=e^{2t}$

$x_0=(C_1+C_2t)e^{2t}$

$x_1=Ae^{2t},\ x_1'=2Ae^{2t},\ x_1''=4Ae^{2t}$

$4Ae^{2t}-8Ae^{2t}+4Ae^{2t}=e^{2t}$

$A=0$

$\therefore x=x_1+x_0=(C_1+C_2t)e^{2t}$

(10) $x''-2x'+x=25\sin2t$

$\lambda^2-2\lambda+1=0$

$(\lambda-1)^2=0\ \therefore\ \lambda=1$ (2重解)

$x_0=(C_1+C_2t)e^{t}$

$x_1=A\cos2t+B\sin2t$

$x_1'=-2A\sin2t+2B\cos2t$

$x_1''=-4A\cos2t-4B\sin2t$

$-4A\cos2t-4B\sin2t+4A\sin2t-4B\cos2t$
$+A\cos2t+B\sin2t=25\sin2t$

$(4A-3B)\sin2t+(4B-3A)\cos2t=25\sin2t$

$\begin{cases}4A-3B=25\\-4B-3A=0\end{cases}\ \therefore\begin{cases}-4B=3A\ \therefore B=-\frac{3}{4}A\\4A-3(-\frac{3}{4}A)=25\end{cases}$

$4A+\frac{9}{4}A=25\ \therefore\ \frac{16}{4}A+\frac{9}{4}A=25$

$\frac{25}{4}A=25\ \therefore\ A=4,\ B=-3$

$\therefore x_1=4\cos2t-3\sin2t$

$x=x_0+x_1$

$=(C_1+C_2t)e^{t}+4\cos2t-3\sin2t$

(11) $x''-x'-2x=12e^{-t}$

$\lambda^2-\lambda-2=0$

$(\lambda+1)(\lambda-2)=0$

$\lambda=-1,2$

$\therefore x_0=C_1e^{-t}+C_2e^{2t}$

$x_1=Ate^{-t}$

$x_1'=Ae^{-t}-Ate^{-t}$

$x_1''=-Ae^{-t}-Ae^{-t}+Ate^{-t}$

$\therefore x_1=-2Ae^{-t}+\cancel{Ate^{-t}}-Ae^{-t}+\cancel{Ate^{-t}}$
$-2\cancel{Ate^{-t}}$

$=-3Ae^{-t}=12e^{-t},\ A=-4$

$x=C_1e^{-t}+C_2e^{2t}-3Ae^{-t}$

$x=x_1+x_0=C_1e^{-t}+C_2e^{2t}-4te^{-t}$

2

(1) $x''=t^2+e^{3t}+\sin5t+\frac{1}{t}$

$x'=\int x''dt=\frac{1}{3}t^3+\frac{1}{3}e^{3t}-\frac{1}{5}\cos5t+\log|t|$

$x=\int x'dt=\frac{1}{12}t^4+\frac{1}{9}e^{3t}-\frac{1}{25}\sin5t+t\log t-t$

$\left(\int\log|t|dt=\underbrace{t\log|t|-\int\frac{1}{t}\cdot t\,dt}_{t\log|t|-t}\right)$

(2) $t^2x''+3tx'-3x=0$

$x=t^\lambda,\ x'=\lambda t^{\lambda-1},\ x''=(\lambda^2-\lambda)t^{\lambda-2}$

$t^2((\lambda^2-\lambda)t^{\lambda-2})+3t(\lambda t^{\lambda-1})-3t^\lambda=0$

$\lambda^2t^\lambda-\lambda t^\lambda+3\lambda t^\lambda-3t^\lambda=0$

$\lambda^2-\lambda+3\lambda-3=0$

$\lambda^2+2\lambda-3=0$

$(\lambda+3)(\lambda-1)=0\ \therefore\ \lambda=-3,1$

$x=C_1e^{-3t}+C_2e^{t}$

(3) $\begin{cases}x'=4x-y+6t\\y'=2x+y+5\end{cases}$

$2x=y'-y-5\ \therefore\ x=\frac{1}{2}y'-\frac{1}{2}y-\frac{5}{2}$

$x'=\frac{1}{2}y''-\frac{1}{2}$

$\frac{1}{2}y''-\frac{1}{2}=2y'-2y-10-y+6t$

$-10+\frac{1}{2}-\frac{20}{2}+\frac{1}{2}=-\frac{19}{2}$

$\frac{1}{2}y''-2y'+3y=6t-\frac{19}{2}$

$\frac{1}{2}\lambda^2-2\lambda+3=0\ \therefore\ \lambda=\frac{1\pm\sqrt{1-4\cdot\frac{3}{2}}}{2}$

$\lambda^2-\lambda+\frac{3}{2}=0$

$=\frac{1\pm\sqrt{-5}}{2}$

$=\frac{1\pm\sqrt{5}i}{2}$

$=\frac{1}{2}\pm\frac{\sqrt{5}}{2}i$

$x_0=e^{\frac{1}{2}t}\left(C_1\cos\frac{\sqrt{5}}{2}t+C_2\sin\frac{\sqrt{5}}{2}t\right)$

$x_1=At^2+Bt+C\ \therefore\ x_1'=2At+B,\ x_1''=2A$

$A-4At-2B+3At^2+3Bt+3C=6t-\frac{19}{2}$

$3At^2+t(3B-4A)+A-2B+3C=6t-\frac{19}{2}$

$\begin{cases}3A=0\\3B-4A=6\\A-2B+3C=\frac{19}{2}\end{cases}\ \therefore\begin{cases}A=0\\B=2\\C=-\frac{19}{24}\end{cases}$

$-4+3C=\frac{19}{2}$

$3C=-\frac{19}{2}$

$C=-\frac{19}{24}$

$\therefore x=e^{\frac{1}{2}t}\left(C_1\cos\frac{\sqrt{5}}{2}t+C_2\sin\frac{\sqrt{5}}{2}t\right)+2t-\frac{19}{24}$

(4) $x''-(x')^2=0$

$x'=p$

$p'-p^2=0$

$\frac{dp}{dt}-p^2=0$

$\frac{dp}{dt}=p^2$

$\frac{p}{dt}\cdot\frac{1}{p^2}=$

$\int\frac{1}{p^2}dp=\int dt$

$-\frac{1}{p}=t+C$

$p=-\frac{1}{t+c}$

$x'=-\frac{1}{t+c}$

$x=\int-\frac{1}{t+c}dt$

$=-\log|t+c|+A$

(5) $xx'+(x')^2+1=0$

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